THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Out of all rectangles with perimeter 32 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 80 **(b)** 64 **(c)** 72 **(d)** 56

2. A rod $3\frac{1}{2}$ m long has $\frac{3}{4}$ m cut off. The remaining length in metres is: 1

(a) $\frac{1}{4}$ **(b)** $4\frac{1}{4}$ **(c)** $2\frac{3}{4}$ **(d)** $2\frac{1}{4}$

3. An isosceles triangle OAB with OA = OB has $\angle AOB = 40^\circ$. Each base angle (degrees) is: 1

(a) 65 **(b)** 80 **(c)** 70 **(d)** 75

4. A regular polygon is rotated again and again through its smallest angle 40°. The number of turns needed to complete a full turn is: 1

(a) 8 **(b)** 9 **(c)** 11 **(d)** 10

5. If ? + (-7) = -12, then the missing number ? is: 1

(a) -7 **(b)** -5 **(c)** 5 **(d)** -3

6. The sides of a triangle are 13 cm, 14 cm and 15 cm. Its perimeter (cm) is: 1

(a) 55 **(b)** 42 **(c)** 43 **(d)** 41

7. To add $\frac{1}{6}$ and $\frac{1}{8}$, the common denominator used is the LCM of 6 and 8. The LCM is: 1

(a) 12 **(b)** 30 **(c)** 48 **(d)** 24

8. The diameter of the circle that exactly fits inside a square of side 10 cm is (cm): 1

(a) 11 **(b)** 10 **(c)** 15 **(d)** 9

9. A regular 6-gon has side 7 cm. Its perimeter (cm) is: 1

(a) 49 **(b)** 42 **(c)** 48 **(d)** 35

10. The value of 7 - 12 - (-7) is: 1

(a) 0 **(b)** 2 **(c)** 12 **(d)** 4

11. A square park of side 20 m has a footpath 1 m wide all around outside it. The area of the footpath (m^2) is: 1

(a) 84 **(b)** 92 **(c)** 80 **(d)** 88

12. $\frac{7}{9} + \frac{2}{3} - \frac{5}{18} =$ 1

(a) $\frac{4}{9}$ **(b)** $\frac{7}{18}$ **(c)** $1\frac{1}{6}$ **(d)** $1\frac{13}{18}$

13. **Assertion (A):** The fraction $\frac{7}{4}$ is an improper fraction. 1
Reason (R): $\frac{7}{4} = 1\frac{3}{4}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A circle has diameter 14 cm. Its radius is 7 cm. 1
Reason (R): The radius of a circle is twice its diameter.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** A rectangle that is not a square has 2 lines of symmetry. 1
Reason (R): Such a rectangle has rotational symmetry of order 2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A carpet 5 m by 4 m is laid in a room 8 m by 6 m. Find the area of the uncovered floor. 2

17. A boy walks $\frac{3}{4}$ km to school and $\frac{2}{5}$ km to a shop, then returns by the same route. Find the total distance. 2

18. A square ABCD of side 10 cm has diagonals meeting at O. Find the area of triangle AOB and the sum of areas of the four triangles. 2

19. A figure has rotational symmetry of order 4. List all the angles (up to but not including $360°$) through which it can be turned to coincide with itself. 2

20. A mountain climber starts at -20 m, goes up 60 m and then down 25 m. Find the climber's final height and its vertical distance from a camp at 30 m. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Write the next three terms of the pattern -20, -14, -8, -2, ... and find the first positive term. 3

22. A rectangular plot 50 m by 30 m is to be fenced with 2 rounds of barbed wire costing ₹15 per metre. Find the total cost. Also find the cost of ploughing it at ₹5 per m^2. 3

23. A school has 360 students. $\frac{4}{9}$ of them are girls. $\frac{3}{8}$ of the girls play chess. How many girls play chess and how many boys are there? 3

24. Four circles of radius 4 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

OR

A designer wants a symmetrical logo with 6 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 6 petals are placed. 3

25. A submarine at -10 m dives 70 m, rises 55 m, then dives 30 m. Find its position after each move and the total vertical distance moved. 3

OR

A square of side 15 cm and a rectangle of length 18 cm and breadth 12 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Anita draws three concentric circles with centre O and radii 3 cm, 5 cm and 8 cm for a rangoli design. 5

(i) What is the diameter of the smallest circle?

(ii) What is the width of the ring between the 5 cm and the 8 cm circles?

(iii) What is the length of the longest chord of the largest circle?

(iv) A point P is 5 cm from O. Say whether P is inside, on or outside each of the three circles.

27. A rectangle with length 24 cm and breadth 10 cm has a triangular piece with base 24 cm and height 6 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A trader sold $\frac{1}{4}$ of his cloth on day 1 and $\frac{2}{3}$ of the remaining on day 2. If 12 metres are left, find the total length of cloth. 5

END OF QUESTION PAPER — Total: 50 Marks

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